Utility and Decision Networks

Lecture 22

CS 4880

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Three Lectures, Three Different Questions

21: State estimation	What might be true about the world right now?
22: Utility and decisions	Given those beliefs, which choice has the best consequences?
23: MDPs and POMDPs	How should we act over time as actions change the world and new observations arrive?

Key idea

Today's job: connect beliefs to preferences and make a defensible choice. Lecture 23 builds policies for sequences of choices.

The Robot Has a Belief, but Still Needs a Decision

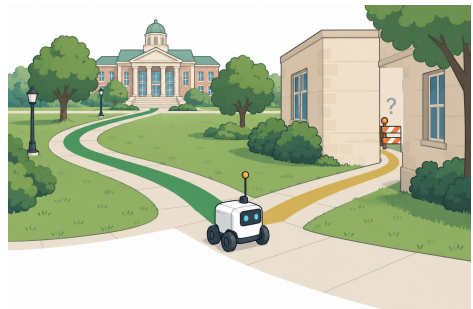
Our campus delivery robot estimates:

- shortcut clear: 65%;
- shortcut blocked: 35%.

It can take the shortcut or a predictable detour.

In class

Which would you choose? What information is missing from the question?



The Most Likely Outcome Does Not Settle the Choice

The shortcut is more likely to be clear than blocked. But we also need to know:

- How much time does a successful shortcut save?
- What happens if the attempt fails?
- How valuable is a predictable arrival?

Probability

How likely is each consequence?

Preference

How desirable is each consequence?

Key idea

An agent needs both a model of what may happen and a model of what matters.

Keep One Decision in View

For this lecture, compare two routes for **one delivery segment**.

- Choosing a route does not change whether the shortcut is currently blocked.
- Each outcome score includes the segment's time, energy, and disruption.
- The detour's outcome is fixed in this simplified model.
- We will also ask whether information is worth buying *before* the route choice.

Key idea

We finish evaluating the example at the end of this segment. Choosing what to do at every later intersection belongs to Lecture 23.

Utility Makes the Tradeoff Explicit

A **utility** is a score for how desirable an outcome is to the agent.

Outcome	Utility	Interpretation
Shortcut succeeds	80	fast, uneventful arrival
Take the detour	45	slower, predictable arrival
Shortcut is blocked	-60	substantial delay and disruption

These are illustrative preference scores, not probabilities or minutes.

Key idea

Higher means preferred. The gaps also matter: they express how much upside compensates for the downside in uncertain choices.

The Scores Are a Modeling Choice

Flexible delivery

A failed shortcut is inconvenient. Saving time may justify more risk.

Time-critical delivery

A failed attempt could miss the delivery window. Reliability may matter more.

The blockage probabilities can be identical while the preferred route differs.

In class

Which outcome score would you reconsider if lateness became much more costly? Why would that change the decision?

Key idea

An accurate probability model cannot decide which outcomes the system should value.

Expected Utility Is a Probability-Weighted Score

Compare the possible consequences of each action using the same utility scale.

$$\text{expected utility} = \sum_{\text{possible outcomes}} \text{probability} \times \text{utility}.$$

In words:

1. List the consequences an action could produce.
2. Weight each consequence's utility by its probability.
3. Add the contributions and compare feasible actions.

Key idea

This is the weighted-average idea from expectimax, using beliefs from our probability model.

Worked Choice: Shortcut or Detour?

Action	Clear: 65%	Blocked: 35%
Shortcut	80	−60
Detour	45	45

$$EU(\text{Shortcut}) = 0.65(80) + 0.35(-60) = 31$$

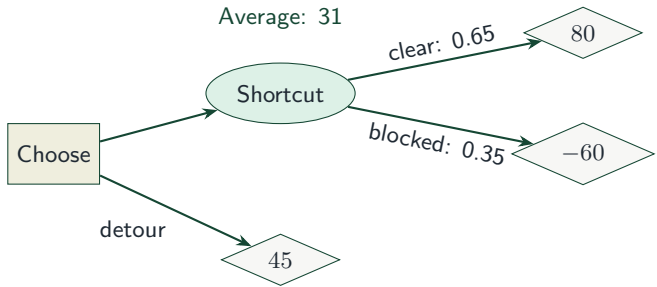
$$EU(\text{Detour}) = 45$$

Here EU means expected utility. Choose the detour: $45 > 31$.

Key idea

The shortcut's likely success does not outweigh its costly failure under these preferences.

The Same Calculation as an Outcome Tree



Key idea

At chance branches, take a weighted average. At a choice, select the action with the higher expected utility.

Better Evidence Can Change the Preferred Action

Keep the same utilities and vary only the chance that the shortcut is clear.

Chance clear	Shortcut EU	Detour EU	Choice
65%	31	45	Detour
75%	45	45	Tie
85%	59	45	Shortcut

In class

Why does “more likely clear than blocked” provide too little information to justify the shortcut?

A Good Decision Can Still Miss the Best Outcome

Suppose the robot takes the detour and later learns the shortcut was clear.

- That does not make the original decision irrational.
- At decision time, the robot had to account for the blocked outcome too.
- We should assess the choice using the information available when it was made.

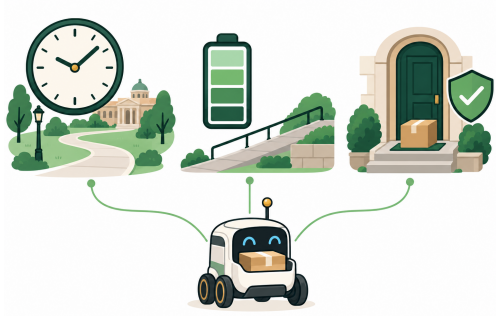
Key idea

Expected utility compares choices before uncertainty is resolved. It does not promise the best result on every individual attempt.

What Should Go Into the Utility Model?

- **Time:** travel time, deadlines, and tolerance for delay.
- **Energy:** battery cost, slopes, and the reserve needed to finish.
- **Reliability:** failed attempts, disruption, and predictable completion.

A weighted score is one option. Its attributes and scales must reflect the intended tradeoffs.



Key idea

Utility design turns an informal goal such as “make a good delivery” into explicit preferences we can inspect and challenge.

Some Requirements Define Which Actions Are Allowed

Before comparing utilities, identify the **feasible actions**.

- A closed walkway can rule out a route entirely.
- Insufficient battery can rule out a long detour.
- A safety envelope can rule out a maneuver near pedestrians.

Then compare the expected utility of the remaining choices.

In class

Should a prohibited route become acceptable merely because it saves enough time? What should the action set represent?

Key idea

Some requirements belong in constraints; other preferences allow tradeoffs within those constraints.

A Decision Network Separates Three Ingredients



What is uncertain?
Is the short-
cut blocked?

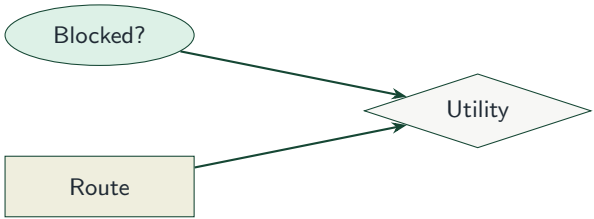
What can we choose?
Shortcut or detour?

What do we prefer?
How good is
the result?

Key idea

A decision network, also called an influence diagram, adds choices and preferences to a probabilistic model.

Our Route Choice as a Decision Network

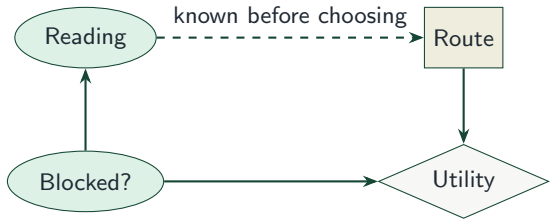


- The score depends on the route and the true blockage state.
- The robot chooses the route; it does not choose the blockage state.
- Blockage is hidden, so the robot averages over its belief.

Key idea

Arrows into utility show what determines the score. The diagram does not make the hidden state observable.

Add a Reading Available Before the Choice



- The sensor reading depends probabilistically on blockage.
- The dashed information arrow says the reading is available to the decision.
- An imperfect reading still leaves uncertainty about blockage.

Use Information the Agent Actually Has

Before a reading

Compare routes using the current 65% / 35% belief.

After a reading

Update the belief, then compare routes again. Different readings can justify different choices.

In class

Why would "choose the shortcut when it is clear and the detour when it is blocked" give the robot an unfair advantage if it has no sensor reading?

Key idea

Average over what remains hidden. A choice may depend only on information available before it is made.

A Scan Is Useful When It Improves the Route Choice

A scan costs time and energy. Its benefit is a better-informed decision.

1. Acquire a reading.
2. Update the belief.
3. Choose a route using that information.

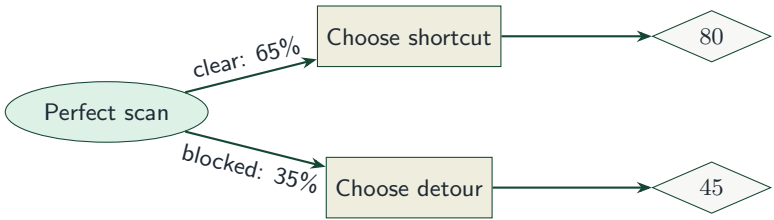
Compare this plan with acting now, using the same outcome utilities and including the scan cost.



Key idea

Value of information asks how much better the decision could become, rather than how much data the sensor produces.

Start with an Ideal Scan That Reveals Blockage Perfectly



Assume the scan leaves blockage and route availability unchanged; its cost captures the delay and energy.

Key idea

We learn first, then choose. Perfect information lets the robot avoid the blocked-shortcut outcome entirely.

How Much Would Perfect Information Be Worth?

Without scanning, the best route has expected utility 45.

$$\text{With perfect information, before cost} = 0.65(80) + 0.35(45) = 67.75.$$

$$\text{Gross value of information} = 67.75 - 45 = 22.75.$$

- If the scan costs 5 utility units: $67.75 - 5 = 62.75$; scan.
- If it costs 25: $67.75 - 25 = 42.75$; take the detour now.

Key idea

Perfect information about blockage is worth at most 22.75 here. A noisy scan of that same variable cannot have greater gross value under the same assumptions.

More Accurate Information Can Still Be Useless for This Choice

Now consider a weak sensor whose readings never make the shortcut preferable to the detour.

- The scan may improve its estimate of blockage.
- The route choice still stays the same.
- Before acquisition cost, the expected improvement in decision utility is zero.

In class

Can you name a measurement that helps describe the world but would not affect this particular route decision?

Key idea

Under these assumptions, free optional information cannot hurt because it can be ignored. Paying to obtain it can still be a poor choice.

What Changes with a Real, Noisy Sensor?

The reasoning stays the same:

1. Consider each reading the sensor might produce.
2. Update the belief for that reading.
3. Choose the best route under the updated belief.
4. Average across readings, then subtract the scan cost.

A noisy reading may leave enough uncertainty that the detour remains best.

Key idea

The important idea is the conditional choice after a reading. The detailed Bayes calculations are in the optional backup slides.

Discussion: Which Scan Would You Buy?

The route, prior belief, and utilities stay as in our example.

Option	Cost	What it tells us
A	5	Perfectly reveals shortcut blockage.
B	25	Perfectly reveals shortcut blockage.
C	1	Improves localization within this intersection, without changing route beliefs, utilities, or feasibility.

In class

Which options improve this route decision after cost? Explain using what the information changes. What would have to change for C to become useful?

The Boundary: One Route Choice Versus a Continuing Strategy

Lecture 22

We score the consequences of a specified route choice, possibly after acquiring information.
We stop at the end of the segment.

Lecture 23

The next state brings new actions and more consequences.
We need a strategy for what to do across those future situations.

Key idea

The distinction is the planning problem we model. Decision networks can also represent multiple decisions; our examples today deliberately isolate one route choice.

Next: Actions Change What Becomes Possible Later

Should the robot recharge before continuing its delivery?

- Charging costs time now but may make later destinations reachable.
- Continuing immediately makes progress now but may leave it stranded later.
- Future readings can change what it should do next.



Key idea

Lecture 23 builds MDPs for sequential decisions with observed states, then POMDPs for sequential decisions with hidden states and beliefs.

Exit Questions

1. Why can two agents with identical blockage beliefs choose different routes?
2. Why can the detour be rational even though the shortcut is probably clear?
3. When does a more accurate sensor have zero gross decision value?
4. What new issue appears when today's action changes tomorrow's available choices?

In class

Answer in words. Identify where probability, preference, information, or sequential consequences enter each answer.

Takeaways: Beliefs, Preferences, and a Choice

- Probability tells us what may happen; utility expresses what matters.
- Expected utility weights each consequence's utility by its probability.
- Decision networks make uncertainty, controllable choices, and utility explicit.
- Information is valuable through the decisions it can improve, after accounting for its cost.
- A good decision uses the information available when it is made.

Key idea

Today: compare consequences for a specified choice. Next: plan how to act over time, with MDPs and POMDPs.

Optional Backup: Formalism and a Noisy Scan

Backup Guide

These slides support questions or further practice; the main lecture ends at the takeaways.

- Compact expected-utility notation and the shortcut threshold.
- What transformations preserve utility-based decisions.
- An additive model for multiple attributes.
- The formal value-of-information comparison.
- A complete imperfect-sensor calculation using Bayes' rule.

Backup: Under Uncertainty, Use Expected Utility

Average the utility of each possible outcome, weighted by its probability:

$$EU(a \mid e) = \sum_{o} P(o \mid a, e) U(o).$$

Then choose among feasible actions:

$$a^* = \arg\max_{a \in \mathcal{A}} EU(a \mid e).$$

Symbols: a: candidate action; e: evidence already observed; o: possible outcome; $P(o \mid a, e)$: outcome probability under the chosen action and evidence. $\sum_o$: sum over outcomes; $\mathcal{A}$: feasible action set; $\arg\max$: return a maximizing action; a^*: chosen action.

Key idea

Probability supplies the weights; utility supplies the values. We will suppress fixed evidence e in the numerical example.

Backup: When Would We Switch to the Shortcut?

Let p be the probability that the shortcut is clear. It beats the detour when

$$\underbrace{80p - 60(1-p)}_{EU(\text{Shortcut})} > \underbrace{45}_{EU(\text{Detour})}.$$

Rearranging gives

$$140p > 105 \qquad \Longrightarrow \qquad p > 0.75.$$

At $p = 0.75$ the actions tie; at our current $p = 0.65$, choose the detour.

Symbols: $p = P(\text{Clear})$; $1 - p = P(\text{Blocked})$; $80, -60, 45$: utilities from the table. EU: expected utility; $>$: strict preference here; $\Longrightarrow$: implies.

Key idea

"More likely clear than blocked" is insufficient. These utilities require more than a 75% chance of a clear shortcut.

Backup: Changing the Utility Scale Preserves the Decision

Under the standard expected-utility consistency assumptions on preferences over lotteries, utilities represent choices under uncertainty.

$$U'(o) = \alpha U(o) + \beta, \qquad \alpha > 0.$$

Then

$$EU'(a) = \alpha EU(a) + \beta.$$

A positive scaling and a common shift preserve which action has the greatest expected utility.

Symbols: o: outcome; a: action; U, U': original/transformed utilities; EU, EU': corresponding expected utilities. α: positive scale; β: common offset. Outcome probabilities sum to 1. A lottery is a probability distribution over outcomes.

Key idea

Arbitrary increasing transformations preserve rankings of certain outcomes, but can change decisions under uncertainty. Utility gaps matter.

Backup: One Engineering Approximation: Multiattribute Utility

Combine attribute scores on stated, comparable scales:

$$U(o) = w_t u_t(o) + w_e u_e(o) + w_s u_s(o) + w_p u_p(o).$$

Symbols: o: outcome; u_t, u_e, u_s, u_p: scores for time, energy, safety, and progress, all oriented so higher is better. w_t, w_e, w_s, w_p: nonnegative tradeoff weights; here subscripts identify attributes, not time steps.

A larger weight matters relative to the score's scale: changing a score's units requires adjusting its weight.

Key idea

An additive model assumes tradeoffs can be separated this way. Attribute interactions may require a richer utility model.

Backup: Value the Plan: Scan, Observe, Then Choose

Compare the best route now with the best route after each possible reading:

$$V_{\text{now}} = \max_a EU(a \mid e), \qquad V_{\text{with}} = \sum_z P(z \mid e) \max_a EU(a \mid e, z).$$

The **gross value of information** is $V_{\text{with}} - V_{\text{now}}$. Acquire the reading when this improvement exceeds its acquisition cost.

Symbols: e: current evidence; a: route choice; z: possible reading; EU: expected utility. V_{now}: optimal value before sensing; V_{with}: expected optimal value after sensing, before its cost. Sum over readings; choose the route after each reading.

Key idea

A scan's immediate cost buys a later decision. Evaluating only its immediate reward would miss its benefit.

Backup: Will a Better Measurement Change the Decision?

The prior probability of blockage is $P(\text{Blocked}) = 0.35$. Without scanning, choose the detour: $V_{\text{now}} = 45$.
A scan costs $c = 5$ utility units and has the sensor model

$$P(+ \mid \text{Blocked}) = 0.90, \qquad P(+ \mid \text{Clear}) = 0.10.$$

Symbols: $+$: sensor reports blockage; $-$: sensor reports clear. These are readings, not true states. c: fixed additive utility cost; V_{now}: value of the best route without another reading.

Assume the world stays fixed during this brief scan, both routes remain available, and its delay and energy cost are captured by c.

Key idea

We are valuing one imperfect measurement before one route choice, using the same utilities as before.

Backup: The Scan Produces Two Possible Belief Updates

Weight each sensor likelihood by its prior state probability:

$$P(+) = 0.90(0.35) + 0.10(0.65) = 0.38, \qquad P(-) = 1 - 0.38 = 0.62.$$

Bayes' rule gives

$$P(\text{Blocked} \mid +) = \frac{0.90(0.35)}{0.38} \approx 0.829,$$

$$P(\text{Blocked} \mid -) = \frac{0.10(0.35)}{0.62} \approx 0.0565.$$

Symbols: $P(+), P(-)$: reading probabilities; $P(\text{Blocked} \mid +)$ and $P(\text{Blocked} \mid -)$: posterior blockage beliefs. The negative-reading likelihood under blockage is $1 - 0.90 = 0.10$; denominators normalize each posterior.

Key idea

Predict how often each reading occurs, then update the belief separately for each branch.

Backup: After the Scan, Choose a Route for Each Reading

For a **positive** reading, blockage probability is about 0.829:

$$EU(\text{Shortcut} \mid +) \approx (1 - 0.829)(80) + 0.829(-60) \approx -36.1.$$

Choose the **detour**, whose utility is 45.

For a **negative** reading, blockage probability is about 0.0565:

$$EU(\text{Shortcut} \mid -) \approx (1 - 0.0565)(80) + 0.0565(-60) \approx 72.1.$$

Choose the **shortcut**, since $72.1 > 45$.

Symbols: $EU(a \mid +)$ and $EU(a \mid -)$: route values after the reading, before sensing cost. $80, -60$: shortcut utilities; 45: detour utility. Displayed posteriors are rounded; calculations use full precision.

Key idea

The conditional plan is: positive $\rightarrow$ detour; negative $\rightarrow$ shortcut.

Backup: Separate Plan Value, Information Value, and Cost

Average the value of the best route in each reading branch:

$$V_{\text{with}} \approx \underbrace{0.38(45)}_{+:\ \text{detour}} + \underbrace{0.62(72.1)}_{-:\ \text{shortcut}} \approx 61.8.$$

Gross information value: $61.8 - 45 = 16.8$.
Scan-plan value after cost: $61.8 - 5 = 56.8$.
Net improvement over acting now: $56.8 - 45 = 11.8$.

Symbols: V_{with}: expected route utility after sensing, before cost. $0.38, 0.62$: reading probabilities; $45, 72.1$: optimized branch values; 5: scan cost. All values use the same utility scale.

Key idea

The noisy-scan plan has value 56.8, a gain of 11.8 over acting now. Its gross information value 16.8 is below the perfect-information bound 22.75.